Predicting Cross-Sectional Relative Returns in the S&P 500: A Feature-Engineering and Walk-Forward Modeling Case Study

Quantitative Research Case Study

1 Problem and Approach

The task is to predict *relative* future stock returns inside the S&P 500: not whether the market goes up, but which stocks will out- or under-perform their peers. This framing is deliberate. Index-level direction is dominated by macro factors that daily OHLCV data cannot forecast, whereas the *cross-section* of returns carries persistent, tradable structure (momentum, reversal, liquidity, volatility, and sector-rotation effects) that is learnable from price and volume alone.

We therefore predict each stock's forward return *in excess of its market and sector*, first test validation-selected model ensembles, and then use the same machinery to build a narrower momentum-plus-residual overlay. The primary trading evaluation is a cost-adjusted, sector-neutral decile long–short portfolio: within each sector, long the top-scored stocks and short the bottom-scored stocks, then combine sectors into a dollar-neutral book. The entire pipeline is built to be honest about look-ahead leakage, overlapping-window inflation, transaction costs, and survivorship bias, because in this problem the easiest way to produce a spectacular-looking result is to make a methodological mistake.

Headline design. Curated technical/liquidity/cross-sectional features → sector- and market-relative forward-return targets → a single-factor momentum baseline → compact Ridge/XGBoost models trained on factor-neutral residual labels → a validation-selected momentum-plus-residual overlay → rank IC / ICIR, DSR/PBO, style-neutral attribution, capacity, and cost-aware portfolio backtests. The research decision is simple: on price/volume plus sector metadata alone, keep low-turnover 12–1 momentum as the research benchmark, specifically the 5-day sector-neutral implementation, but treat it as a signal benchmark rather than a fundable standalone product. Its case rests on traded-horizon IC and turnover, not on a tight portfolio Sharpe: block-bootstrap IC $t = 2.41$ with mean-IC CI $[0.004, 0.042]$, turnover 0.22, net Sharpe 0.62–0.63, mean-return $t = 1.31$, and bootstrap Sharpe CI $[-0.32, 1.63]$. A calendar-slice stress test shows that the recent hold-out is momentum-friendly rather than representative. A partial inclusion-side PIT filter cuts the horizon-matched Sharpe from 0.620 to 0.461 and its own bootstrapped Sharpe CI is $[-0.49, 1.43]$, so the portfolio Sharpe evidence is directional only. I would size any paper exposure off IC stability and turnover, not this recent-regime Sharpe; the kill criterion is a true point-in-time universe with delisting returns plus CPCV-style multi-regime stability. Reported net returns include explicit trading costs but exclude stock borrow; a simple 25/50/100 bps annual short-borrow haircut on 50% short notional reduces the 5-day momentum Sharpe from 0.627 to 0.614/0.600/0.574. The ML/graph/residual models are not promoted as standalone alpha: the 30-day residual overlay is a horizon-specific sleeve, not evidence that ML improves the endorsed 5-day benchmark. Over 35 non-overlapping periods its paired return increment has $t = 0.25$, 93% correlation with momentum, a Sharpe-gap CI crossing zero, full-grid DSR 0.003, and PBO 50%. The next research dollar should therefore go to a true point-in-time universe with delisting returns and lower-turnover execution tests, not to a larger model zoo.

Momentum's status as the core research benchmark is anchored first on the traded-horizon evidence: the 5-day sector-neutral signal has positive block-bootstrap IC support ($t = 2.41$, mean-IC CI $[0.004, 0.042]$) at the lowest turnover among tested horizons. The conclusion is not a post-hoc claim that 5d was the single most significant horizon: all tested traded horizons have positive mean IC, and 5d is selected for the combination of positive block-bootstrap IC support, lower turnover, and PIT robustness. The pre-registered 1-day cross-sectional IC confirmation (plain $t = 3.54$; Newey–West lag-21 $t = 4.39$) is corroborating evidence that the signal exists, not the trading statistic. At longer traded horizons the same signal is lower-power: a focused HAC/block check gives hold-out block-bootstrap t of 2.13/1.89/1.85 at 10/20/30 days, with the 20d/30d block-bootstrap IC intervals crossing zero. The standalone 20-day momentum book has a higher Sharpe point estimate (0.83), but its IC interval crosses zero, its portfolio Sharpe is still imprecise (mean-return $t = 1.69$, bootstrap CI $[-0.12, 1.90]$), and the PIT inclusion filter cuts Sharpe from 0.828 to 0.513 with a filtered CI of $[-0.45, 1.58]$. The standalone 30-day momentum book has Sharpe 0.740, mean-return $t = 1.51$, an i.i.d. bootstrap Sharpe interval of $[-0.22, 1.85]$, and a PIT inclusion haircut from 0.741 to 0.446 with a filtered CI of $[-0.52, 1.50]$.

2 Data

We use only two files: `sp500_stocks.csv` (daily OHLCV per ticker) and `sp500_companies.csv` (sector, sub-industry, headquarters, founding year, and index `date_added`). No fundamentals, analyst, macro, or external listing data are used. The cleaned panel covers 503 symbols from 2000-01 to 2026-06 (2.926M stock-day rows).

Price adjustment (data quality). The raw file exposes only a `close` column and no separate `adj_close`. We therefore audited major split events and verified that this `close` behaves as an adjusted close price: around NVDA's 2024-06-10 10:1 split (120.68 → 121.58) and AAPL's 2020-08-31 4:1 split (121.06 → 125.17) there is no split-day gap, consistent with a `yfinance auto_adjust=True` export. In this study, `close` is therefore treated as the adjusted close used for all returns, momentum, volatility, and forward-return targets. Residual uncertainty about dividend adjustment is treated as a minor data-quality caveat.

Raw cleaning. The CSV-to-Parquet preparation step normalizes ticker strings, enforces one row per `symbol,date`, drops rows with missing core OHLCV fields, non-positive prices or volume, and impossible OHLC relationships (`high` below open/low/close or `low` above open/high/close). The raw stock file has no core missing values or duplicate stock-days, but it does contain 4,413 zero-volume rows and one OHLC inversion; these 4,414 rows are removed before feature construction. Large but plausible moves are audited rather than deleted: 44 stock-days have absolute one-day returns above 50%, while none exceed the 300% hard error threshold. Company metadata are also normalized and deduplicated; all 503 tickers have sector and sub-industry labels.

Table 1: Single-factor sanity check on core features, target `target_excess_sector_fwd_5d`. Mean IC is the average daily Spearman rank correlation after the production cleaning/evaluation controls.

Feature	Mean IC	Interpretation
`sector_rank_sma_gap_20d`	-0.0131	within-sector short-term reversal
`ret_5d`	-0.0125	recent winners partially mean-revert
`amihud_20d`	+0.0123	liquidity / price-impact information
`log_dollar_volume`	-0.0123	liquidity/size relation in relative returns
`mom_252d_skip_21d`	+0.0102	12–1 month momentum survives weakly
`rsi_14d`	-0.0099	overbought names tend to cool off

Survivorship bias. The constituent list is effectively the *current* membership applied to history, so firms that went bankrupt, were delisted, or were acquired (Enron, Lehman, Bear Stearns, WaMu, Sears, Kodak, Yahoo, Celgene, Xilinx, . . .) are absent. This is a major current-membership bias. Its magnitude and net sign are unidentified without deleted-name holdings and delisting returns: inclusion-side bias inflates later entrants, while deleted bankrupt losers would have been profitable shorts. The metadata include `date_added`, which supports a partial point-in-time inclusion check: 72 current members were added on or after the 2022 hold-out start. We use that field as a sensitivity in §6, but still do not claim a true survivorship correction because deleted names and delisting returns are missing.

3 Feature Engineering

Each row is one stock-day observation, and every feature uses only information available at or before that date (all rolling windows are backward-looking with an explicit `shift`). We build ≈500 features in economically motivated groups:

- **Price / momentum / reversal:** multi-horizon returns (1–252d), log returns, skip-window momentum, momentum acceleration, short-term reversal, intraday/overnight decomposition, close-in-bar location. *Captures short-term reversal and medium-term momentum.*
- **Volatility / tail / drawdown:** rolling, upside/downside, Parkinson and Garman–Klass volatility, ATR, skew/kurtosis, drawdown from recent highs. *Controls for risk state and crash sensitivity.*
- **Trend / technical:** SMA/EMA gaps and slopes, MACD, RSI, Bollinger z-score/bandwidth/%b, channel position, breakout distance, Williams %R, CCI. *Encodes overbought/oversold and trend regimes.*
- **Liquidity / volume:** log dollar volume, volume z-scores and momentum, Amihud illiquidity, return-per-dollar-volume. *Captures attention, liquidity shocks, and transaction-cost pressure.*
- **Market regime:** equal-weight market return, breadth, cross-sectional dispersion, market volatility. *Relative-return predictability is regime-dependent.*
- **Statistical linkage / residual risk:** rolling market beta, market correlation, idiosyncratic return, and idiosyncratic volatility. *Captures stock-level residual behavior after market exposure is removed.*
- **Sector / sub-industry and peer / style / geography:** group returns, breadth, dispersion, leave-one-out peer means, headquarters-region peers, and style buckets. *Separates stock-specific alpha from group movement.*
- **Graph relation features:** two graph feature families are kept separate. The original `fixed_graph` variant uses deterministic same-sector/sub-industry, style-nearest-neighbor, and rolling-correlation graph relation embeddings. The `supervised_graph` variant uses out-of-fold supervised GAT embeddings and scores. *This separates static peer-structure features from supervised relation-aware graph learning.*
- **Cross-sectional transforms:** date-wise z-scores/ranks plus sector- and sub-industry-neutral versions of the key signals. *Because the target is relative, cross-sectional normalization is usually more informative than raw levels.*

Feature audit and sanity check. The rebuilt manifest contains 525 non-target columns and 40 target columns, but the default walk-forward models use a curated 61-column `core` set covering momentum/reversal, volatility, technical indicators, liquidity, market/industry context, and key cross-sectional ranks. The larger grid treats deterministic peer-graph features and out-of-fold supervised-GAT embeddings as feature variants only; all variants use the same targets, folds, models, costs, purge/embargo, and hold-out protocol, and graph learning earns no separate alpha claim below.

We explicitly verified that the expected factor families are present: 12-minus-1-month momentum (`mom_252d_skip_21d`), short-term reversal (`rev_*`), volatility/range estimators, Bollinger features, `MACD`, RSI-style oscillators, Amihud illiquidity, and date/sector/sub-industry z-score and rank transforms. The short-cycle feature block also includes the signals most likely to be orthogonal to 12–1 momentum: 1–5 day reversal, Amihud liquidity/impact, volume shocks, market breadth, cross-sectional dispersion, and rolling idiosyncratic return after market-beta removal. As a usefulness check, we ran a single-factor daily rank-IC scan on the cleaned 2008–2026 panel (after long rolling-window warm-up for 12–1 momentum and linkage features) using the same T+1 label shift, bottom-10% liquidity filter, and 1%/99% within-date winsorization as the modeling pipeline. Table 1 shows the largest effects in the core set. The magnitudes are small, as expected for daily cross-sectional equity alpha, but the signs are economically coherent: short-term overextension tends to reverse, 12–1 momentum is positive, and liquidity/impact variables contain weak but measurable information.

Prediction target. For horizon h we define the forward total return $r_{i,t}^{(h)}$ and subtract a contemporaneous group mean to obtain sector/market-relative targets, e.g. $\text{target\_excess\_sector\_fwd\_}h\text{d}_{i,t} = r_{i,t}^{(h)} - \bar{r}_{\text{sector}(i),t}^{(h)}$. Targets are forward-looking and are strictly excluded from the feature matrix. The target generator also writes a date-wise percentile-rank target, `target_rank_fwd_hd`, so short-horizon experiments can use volatility-normalized, ranked, or winsorized labels when raw returns are too noisy. The implemented horizon grid covers $h \in \{1, 5, 10, 20, 30, 40, 50, 60, 70, 80, 90\}$. The original broad search emphasized 5d/10d, but the final Route B rerun uses the 30d sector-relative horizon because the shorter horizons did not deliver reliable net economics after costs. This is a selection step, not an exogenous design fact, and is one reason the final overlay must pay the full multiple-testing charge.

Execution timing. The raw target columns are generated on each feature date, but the model is evaluated with a one-day implementation lag. A feature row at date t is interpreted as information available after the t close; the signal is therefore tradeable only from $t{+}1$. In code, the effective training label is the raw target shifted one trading day forward, so a row dated t is matched to the forward return window that starts from the next trading day. This is the convention used for both rank-IC evaluation and portfolio PnL.

4 Methodology

4.1 Models

We compare, from simple to flexible: (i) a **single-factor momentum** baseline (rank by 12-minus-1-month momentum); (ii) **Ridge** regression (linear, strong regularization); (iii) **LightGBM** and (iv) **XGBoost** gradient-boosted trees (non-linear, interaction-aware); and (v) a small **PyTorch MLP** (two hidden layers, dropout, weight decay, early stopping on validation). The MLP is kept deliberately small and is *not* tuned until it wins; §5 reports its honest standing — competitive on in-sample validation IC, but not stable enough to headline once the single-evaluation hold-out, turnover, and momentum benchmark are applied.

Momentum-residual overlay. Because the single-factor momentum baseline is strong, the implemented post-training evaluator does not force ML to replace momentum. For each date, both the baseline score and model score are converted to cross-sectional z-scores, the model score is residualized on the momentum score, and the final overlay signal is

$$s_{i,t} = z(\text{momentum}_{i,t}) + \lambda z(\widehat{\varepsilon}_{i,t}),$$

where $\widehat{\varepsilon}_{i,t}$ is the model component orthogonal to momentum within that day's cross-section. The overlay weight is selected from a small pre-registered grid on validation net portfolio metrics, with $\lambda{=}0$ reproducing the momentum baseline. The final Route B run selects $\lambda{=}0.2$ on validation Sharpe; the test split is not used to tune the amount of residual ML alpha.

Route B residual-target training. After the broad model/graph search failed to produce a robust replacement for momentum, I reran the main experiment with a narrower design: each effective forward label is residualized cross-sectionally by date against sector dummies and OHLCV style proxies (liquidity, volatility, beta/idiosyncratic volatility, short-term reversal, 12–1 momentum, overnight/intraday return, volume shock, breadth and dispersion). Ridge/XGBoost then train on this residual target. The final tradable signal is still evaluated on raw forward returns and is combined with the momentum baseline as a small residual sleeve rather than a replacement.

Graph feature variants. Graph features are predictive inputs, not separate trading rules. The grid distinguishes deterministic `fixed_graph` peer embeddings from out-of-fold supervised-GAT embeddings and their combined variant, all under the same purge, embargo, execution-lag, validation, cost, and portfolio rules. The results below treat graph learning as an ablation family only: it does not earn a headline alpha claim.

4.2 Validation-selected rank ensemble

The ensemble is the final *prediction layer*, not a portfolio construction rule. For each base model m and date t, raw predictions are converted into a cross-sectional percentile rank $q^{(m)}_{i,t} = \text{rank}_t(\hat{y}^{(m)}_{i,t})$. The final score is then an average of member ranks,

$$\text{final\_score}_{i,t} = \sum_{m \in \mathcal{M}} w_m q^{(m)}_{i,t},$$

with either equal weights or non-negative weights proportional to validation rank ICIR. Member selection uses validation rank ICIR and stability gates rather than validation mean IC; the post-validation test split is never used to select models, tune hyperparameters, choose ensemble members, or choose weights.

We compare five ensemble definitions:

- `ensemble_all_val_selected`: the current full ensemble, using the validation-selected best run for each model and feature variant.
- `ensemble_no_mlp`: the same rule after excluding all MLP members.
- `ensemble_tree_only`: Ridge, LightGBM, and XGBoost members only.
- `ensemble_supervised_graph_tree`: only supervised-graph Ridge, LightGBM, and XGBoost members.
- `ensemble_stability_filtered`: only members with positive validation rank IC, enough positive validation years, and non-negative available validation-regime rank IC.

This is also why the robustness tables report `ensemble_no_mlp` and `ensemble_tree_only`: the triage runs show that MLPs can look good on validation IC but are not stable enough to headline unless they pass the same walk-forward and regime gates.

4.3 Leakage controls

Every result below is produced under the same defenses:

- **One-day execution lag:** scores computed on day t are traded at $t+1$, so no same-bar information is used.
- **Label purge:** training rows whose forward label window crosses a split boundary are dropped, on both the training and the evaluation side.
- **Horizon embargo:** the validation/test window starts an h-day embargo after training ends, removing label overlap at the seam.
- **Train-only fitting:** imputation medians, scaling statistics, and the model are fit on training rows only.
- **Universe & winsorization:** before splitting, each date's cross-section drops the bottom 10% by dollar volume and clips features at the 1%/99% within-date quantiles, so illiquid names and outliers do not drive the fit or an untradeable backtest.

4.4 Avoiding overlapping-window inflation

A subtle but decisive issue: if an h-day forward return is booked every day, consecutive observations overlap by $h-1$ days, their IC series is strongly autocorrelated, and a naive ICIR $= \sqrt{252} \cdot \overline{\mathrm{IC}}/\sigma(\mathrm{IC})$ together with an overlap-booked Sharpe explode with h — a pure artifact, not alpha. We fix this in two ways: (1) the portfolio rebalances every h days so booked returns are *non-overlapping*; and (2) ICIR is annualized from a non-overlapping IC subsample (one observation per h days, scaled by $\sqrt{252/h}$). We retain the naive value as `rank_ic_ir_raw` only to document the inflation (Fig. 5). A sanity gate flags any run with Sharpe>3 or ICIR>5 as suspected leakage/overlap.

4.5 Validation: walk-forward selection + hold-out

A single static split can land in a lucky regime. The single-model horizon comparison therefore uses **expanding yearly walk-forward**: train on all data up to year Y, validate on year $Y+1$ (with purge + embargo), then roll forward one year and expand the training window. Model selection uses *walk-forward validation ICIR* and fold stability, not validation mean IC. The fixed 61-feature core is used instead of the full 525-column catalog for the same reason: it retains the economically interpretable, stable feature families while avoiding a high-variance short-horizon fit. The larger supervised-graph ensemble grid uses the same leakage controls but a fixed research split for tractability: train through 2018, validate through 2021, and evaluate the post-validation 2022–2026 test split once. In both cases, test results never participate in feature, model, hyperparameter, ensemble-member, or weight selection.

4.6 Portfolio and metrics

On each rebalance date we sort stocks by the reported score, either the validation-selected ensemble from §4.2 or the final Route B momentum-plus-residual overlay. Portfolio construction is deliberately written as a ladder from simple to stringent:

- **Long-only top bucket:** hold the top decile, either equal-weighted or score-weighted, with no short book. This is closest to a constrained long-only product and is evaluated with annualized return, volatility, Sharpe, max drawdown, turnover, and cost-adjusted return.
- **Plain long–short decile:** long the top 10%, short the bottom 10%, with equal dollar capital on both sides. This is the standard alpha ranking test corresponding to rank IC.
- **Sector-neutral long–short decile:** rank stocks within each sector, build top-minus-bottom decile books inside usable sectors, and average sector books into the total portfolio. This is the primary portfolio in the reported ensemble backtest because the target itself is sector-relative.
- **Market/sector beta-neutral portfolio:** an institutional-style robustness extension that constrains portfolio beta and sector exposures near zero in addition to dollar neutrality. It is stricter than the decile rule, but requires an optimizer and typically reduces raw return.
- **Cost-aware / turnover-controlled variants:** deduct explicit transaction costs, rebalance every 5/10/20 trading days or every target horizon, smooth scores with EWMA, and optionally restrict rebalancing to material signal changes through a no-trade band. The residual-turnover evaluator also sweeps no-trade bands of 0/5/10/15% of book turnover, which targets the gap between model turnover ($\sim$0.55–0.70 per rebalance) and the slower momentum benchmark ($\sim$0.16).

The empirical tables below use the sector-neutral long–short decile as the primary portfolio, rebalanced every h days with a 5 bps per-side transaction cost applied to realized turnover. We report rank IC, overlap-adjusted ICIR, decile spread, annualized net return, volatility, net Sharpe, max drawdown, and turnover. These net returns are gross of stock-borrow and financing costs unless explicitly stated; §6 adds a short-borrow sensitivity. Report tables use arithmetic annualized net return ($\overline{r}_{\mathrm{period}} \times 252/h$) to match the main evaluator; some archived paired-increment summaries also store geometric annualized returns for the same return streams, so return percentages can differ slightly while Sharpe and drawdown tie out.

4.7 Frequency-band and sleeve-book diagnostics

The final diagnostic pass asks whether a product should remain model-light: instead of treating LGBM/XGBoost as the center of the system, test whether simple economic sleeves survive their own native horizons and whether ML earns a small, explicitly deflated role. LGBM is therefore not needed for the core factor book; it is only a falsifiable add-on for regime-conditional sleeve weights, within-band residual combination, or a shrinkage portfolio optimizer. Each of those roles is charged with the same DSR/CPCV skepticism as the model grid, because the available regime episodes are few and the factor-level Fama–MacBeth evidence is weak.

The signal-frequency diagnostic pre-registers 12 single factors and evaluates them against sector-relative labels at horizons 1, 5, 20, 40, 60, and 90 days. It reuses the production T+1 execution lag, bottom 10% liquidity filter, 1%/99% within-date winsorization, and overlap-adjusted ICIR. For each signal, the native horizon is selected by maximum absolute discovery ICIR using data through

2021; the 2022+ hold-out confirms or rejects that band. ICs are computed by re-ranking both signal and target on the common non-missing cross-section for each date.

The two-sleeve book then uses only the signals that survived the hold-out sign screen: 12–1 momentum and short reversal. For each sleeve k and rebalance date t, sector-neutral decile weights $w_{i,t}^{(k)}$ are formed from the signed raw signal. Combined weights are constructed at the position level, not by blending return streams,

$$w_{i,t}^{\text{book}} = \sum_k \alpha_k w_{i,t}^{(k)}, \qquad \alpha_k = \frac{1/\widehat{\sigma}_{k,\text{disc}}}{\sum_j 1/\widehat{\sigma}_{j,\text{disc}}},$$

where $\widehat{\sigma}_{k,\text{disc}}$ is the discovery-period net-return volatility. Because each sleeve is dollar-neutral, the combined book remains dollar-neutral. Costs are charged from the actual combined position turnover $\tau_t = \frac{1}{2}\sum_i |w_{i,t}^{\text{book}} - w_{i,t-1}^{\text{book}}|$.

For the multiple-testing check, the combined book's hold-out Sharpe is deflated in per-observation Sharpe units. With per-rebalance Sharpe SR, skew γ_3, kurtosis γ_4, and T observations, the PSR denominator is

$$d = \sqrt{1 - \gamma_3 SR + \frac{\gamma_4 - 1}{4} SR^2}.$$

The expected maximum null Sharpe for N searched single-factor trials is

$$SR^* = \sigma_{SR}\left[(1-c_E)\Phi^{-1}\left(1-\frac{1}{N}\right) + c_E\Phi^{-1}\left(1-\frac{1}{Ne}\right)\right],$$

where $c_E = 0.5772$ is Euler's constant and σ_{SR} is the cross-trial standard deviation of per-observation Sharpes. The reported DSR is

$$\Phi\left(\frac{(SR - SR^*)\sqrt{T-1}}{d}\right),$$

with $N = 12$ for the pre-registered single-factor scan. The Route B full-grid DSR uses the same formula with $N = 884$, counting the archived searched model/target/horizon/configuration rows in the mirrored walk-forward and residual-overlay experiment grid; this is intentionally conservative because the rows pool heterogeneous horizons and target families.

5 Results

Validation selection. Selecting by across-fold mean validation ICIR, the small MLP is the strongest model at the 1-, 5-, and 30-day horizons (e.g. sector-relative 5d ICIR 1.15, market-relative 0.99) and LightGBM at 20 days (market-relative 0.99); Ridge and XGBoost trail (Fig. 4). No single model dominates every horizon and the boosted trees do *not* uniformly win — the MLP is genuinely competitive on the in-sample IC metric, so we do not pretend trees are categorically better.

Ensemble and graph comparison. The 5-day rank ensembles do not change the investment conclusion. Excluding MLP improves the post-validation rank IC relative to the full ensemble, and the supervised-graph tree ensemble is the strongest 5-day ranker among the archived ensemble diagnostics, but net Sharpe is only about zero to 0.05 after costs and turnover remains near 0.7 per rebalance. I therefore keep ensemble/GAT results as negative model-selection evidence rather than as a tradable alpha claim.

The decisive test: the single-evaluation hold-out. The validation ranking does not survive out of sample. On the 2022–2026 hold-out the *single-factor momentum baseline* (rank by 12-minus-1-month momentum, no fitting) beats every trained multivariate model at the original 5-day benchmark horizon on rank IC (0.023 vs $\leq$ 0.013), ICIR (1.02 vs $\leq$ 0.73), and turnover (0.22 vs $\sim$ 0.59). Its corrected 5-day Sharpe is 0.63, not the legacy $>$ 1 figure, so the honest claim is narrower: momentum is the cleaner rank signal and the lower-turnover book, while Sharpe leadership depends on horizon and evaluator convention (Table 5). The models' validation-IC edge is therefore largely overfit: a robust, low-turnover factor is hard to beat after costs. This is the central, deliberately humbling finding of the study.

Signal-frequency diagnostic. The single-factor frequency scan provides an independent check on that conclusion. The discovery-period ICIR surface shows economically sensible bands: short reversal signals are negative at 1–5 days, low-risk variables flip sign from short to longer horizons, and Amihud illiquidity is the discovery-period champion across 20–60 days, but the 2022+ confirmation step is much harsher (Table 2). At its native 1-day confirmation horizon, momentum is the only signal with a positive, significant, and stronger hold-out t-stat. Amihud's in-sample strength collapses from $t = 5.51$ to $t = 0.41$, volume shock flips sign, and reversal keeps the expected sign but is weak. A follow-up traded-horizon momentum check using `report/artifacts/momentum_ic_traded_horizons/` keeps the horizon distinction explicit: block-bootstrap t is 2.41 at 5d, 2.13 at 10d, 1.89 at 20d, and 1.85 at 30d; the corresponding block-bootstrap mean-IC intervals are positive at 5d/10d but cross zero at 20d/30d. This is a clean warning that 1-day IC confirms signal existence, while tradability is horizon- and cost-limited even before adding ML. The daily Newey–West t-stat rising from the plain $t = 3.54$ to lag-21 $t = 4.39$ is not a HAC sign error; the daily IC series is mildly negatively autocorrelated, so the Bartlett long-run variance estimate tightens rather than widens.

Table 2: Pre-registered signal-frequency diagnostic. Native horizon is chosen by discovery-period absolute ICIR through 2021, then confirmed on the 2022+ hold-out under the production T+1, liquidity, winsorization, and overlap-adjusted ICIR rules.

Signal	Native h	Disc. ICIR	Disc. t	Test t	Hold-out read-through
`mom_252d_skip_21d`	1d	0.595	2.23	3.54	Signal survives; stronger OOS.
`amihud_20d`	40d	1.467	5.51	0.41	In-sample champion, OOS decay.
`ret_5d` reversal	1d	-1.027	-3.84	-0.94	Sign stable, but too weak.
Low-risk block	1d	-0.432 to -0.793	-1.62 to -2.97	≈ 0	Vol/idiosyncratic-vol/beta effect decays.
`volume_z_20d`	60d	0.514	1.93	-1.93	Sign flips on the hold-out.

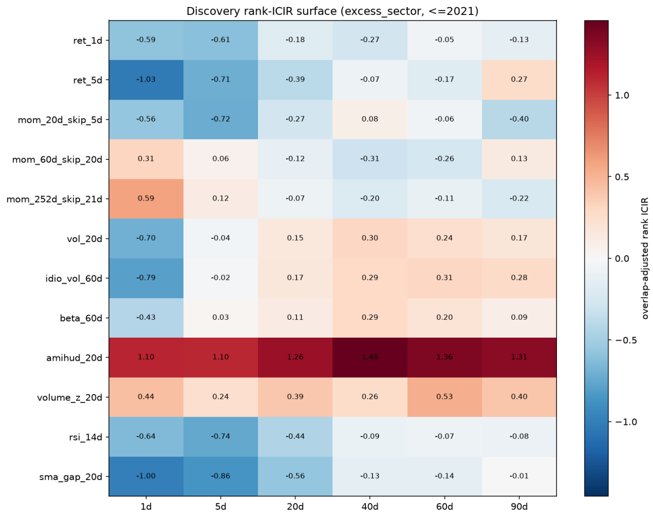

Figure 1: Discovery signal × horizon ICIR surface for the pre-registered single-factor scan.

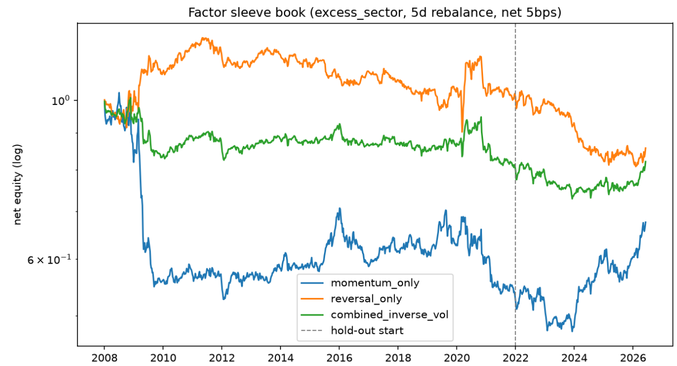

Figure 2: Net equity curves for the 5-day momentum/reversal sleeve-book diagnostic.

Route B 30-day residual overlay. The secondary residual-sleeve lead comes from the narrower residual-target rerun, not from the graph/MLP zoo. XGBoost is trained on 30-day sector-relative labels after daily ridge residualization against sector and OHLCV style proxies. The tradable score is then $z(\text{momentum}) + 0.2z(\text{residual XGBoost})$, with $\lambda = 0.2$ selected on validation Sharpe. Table 3 compares this Route B overlay with standalone momentum and the strongest plain 30d sector model on the 2022–2026 hold-out. This residual-label XGBoost rerun is separate from the broad-grid 30d sector MLP row in Table 5. It is a deliberately separate 30-day sleeve diagnostic, not an overlay on the endorsed 5-day benchmark. The longer horizon is where residual structure and turnover looked most plausible; a 5-day Route B overlay remains untested. The overlay improves the point-estimate economic profile relative to momentum (Sharpe 0.74 → 0.99, max drawdown −13.8% → −7.3%), but a paired increment test shows that this is mostly volatility and drawdown dampening rather than new return: mean per-period return improves by only 0.06% (about 0.5% annualized), the paired return-difference t-stat is 0.25, the overlay–momentum correlation is 0.93, and a paired bootstrap Sharpe-gap CI is $[-0.17, 0.63]$. The evidence is therefore deliberately read conservatively: the sample has only $N = 35$ non-overlapping periods, the absolute bootstrap Sharpe CI is $[0.019, 1.796]$, full-grid DSR is 0.003 after 884 archived searched model/target/horizon/configuration trials, PBO is 50%, and the Rank IC change is within evaluator noise. The standalone momentum baseline is also thin-sample as a 30-day portfolio: Sharpe 0.740 has mean-return $t = 1.51$ and an i.i.d. bootstrap Sharpe interval of $[-0.22, 1.85]$ over the same 35 periods. The robustness verdict is one-sided: Route B is a drawdown/risk sleeve worth one more targeted audit, not deflated alpha and not a replacement for 5-day momentum. The plain 30d LightGBM row shows why horizon framing matters: a raw model can beat the overlay's Sharpe point estimate, but it fails the paper's broad-grid walk-forward validation-selection rule (mean validation RankICIR 0.0804 versus the selected MLP's 0.4882 on that selection scale), sits below the 884-row expected-max null Sharpe of 1.512, and was not part of the archived Route B DSR/PBO audit. I therefore do not use it to claim deflated alpha or to tune the final story after seeing the hold-out. The overlay is kept only for its economic sleeve role and lower turnover, not for raw performance dominance. `report/artifacts/route_b_cost_breakeven/` recomputes both 30-day books from the same return/turnover streams at alternative per-side costs: the overlay Sharpe gap remains 0.246 at 5 bps, 0.212 at 25 bps, and 0.169 at 50 bps, with a Sharpe break-even near 147 bps per side. This means ordinary commission/slippage variation is not the main Route B objection; the blocker is still the weak paired return increment and high momentum correlation.

Table 3: 30-day sector-relative hold-out comparison, net of 5 bps/side cost and gross of stock-borrow. Route B has only $N = 35$ non-overlapping periods: momentum Sharpe CI $[-0.22, 1.85]$, overlay CI $[0.019, 1.796]$, full-grid DSR 0.003 over 884 searched trials, and CSCV PBO 50%. It is a candidate residual sleeve, not a momentum replacement; paired returns are archived in `report/artifacts/route_b_paired_increment/`.

Config	Source	Rank IC	ICIR	Net Sharpe	Ann. Ret	Max DD	Turnover
Momentum baseline	canonical baseline	0.0282	0.698	0.740	7.78%	-13.83%	**0.401**
Plain LightGBM model	raw hold-out diagnostic; val-rejected/unaudited	0.0362	0.802	1.226	12.95%	-8.16%	0.631
Route B overlay, $\lambda = 0.2$	val-selected sleeve	0.0279	0.759	0.987	8.27%	**-7.32%**	0.494
Paired increment	overlay − momentum	–	–	Δ0.246	Δ0.50%	+6.51pp	Δ0.093
Route B overlay, $\lambda = 0.3$	diagnostic only	0.0280	0.806	1.201	9.98%	-6.10%	0.521

Route B artifact check. Local artifacts reproduce the canonical 30d momentum row exactly (Sharpe 0.740, annual return 7.78%, max drawdown −13.83%, turnover 0.401) and recompute the selected overlay against the same raw label: canonical Rank IC is 0.0279 versus momentum's 0.0282, with ICIR 0.759 versus 0.698. The archived paired return stream is the cleaner economic comparison: across the same 35 rebalance dates, period volatility falls from 3.63% to 2.89%, but mean return improves by only 0.059% per period and the paired return-difference t-stat is 0.25. This reframes Route B as a high-correlation risk-sleeve candidate, not independent residual alpha. A shifted six-rebalance inverse-vol control on the same 30-day momentum returns does not reproduce the drawdown result: aligned Sharpe is 0.73 with max drawdown −14.0%, versus Route B's 0.92 and −7.3% (`report/artifacts/route_b_vol_timing_control/`). This is a coarse return-stream negative control, not an executable capacity test.

Momentum/reversal sleeve-book diagnostic. The next product-style check built the two-sleeve book implied by the frequency scan: momentum plus short reversal, rebalanced every 5 days and combined at the position level with discovery inverse-volatility weights. The estimated discovery weights are 43.1% momentum and 56.9% reversal, and the full-sample sleeve net-return correlation is mildly diversifying (-0.153), but the hold-out economics do not justify a second sleeve: momentum-only has the best 2022+ Sharpe (0.598), reversal-only is negative (-0.519), and the combined inverse-vol book has Sharpe 0.094 with DSR 0.055 after the 12-factor scan and bootstrap Sharpe interval $[-0.724, 1.095]$. This uses raw signed factor sleeves rather than the rank-transformed baseline in Table 5, so the Sharpe levels are not apples-to-apples; the useful conclusion is that a simple multi-sleeve product does not beat momentum in this dataset.

The full sleeve table is omitted for space. An earlier 10-day residual overlay improved point-estimate Sharpe from 0.746 to 0.995, but the later Route B audit shows why such overlays remain exploratory without deflation and attribution support.

Graph relation ablation. The completed 120-candidate tree grid compares real relation families (sector, style-nearest-neighbor, rolling correlation, and all relations) against random-edge and no-edge controls for 5d/10d sector- and market-relative targets. The result is negative enough to summarize rather than headline: placebo controls beat real relations in three of four target blocks, and the one narrow real-relation win remains below the momentum ICIR. Graph learning is therefore treated as an experimental feature family, not a source of reported alpha.

Failed ablations and negative controls. The negative controls remain part of the research record but are compressed here because they are not alpha evidence: a 24-member fixed-graph tree seed ensemble reached only Rank IC 0.0171, ICIR 0.517, Sharpe 0.290, and turnover 0.649, while the best Kaggle/Ubiquant-style date-aggregate sector 5d test ICIR was 0.569, below the momentum ICIR of 1.016. These failures are important because they prevent the paper from presenting graph features, seed bagging, or date aggregates as successful complexity.

PM-level robustness audit. The Route B 30-day overlay is the best current residual-sleeve research lead, but it is still *not* a production-grade alpha claim. A post-training audit treats the full search as multiple testing, applies CSCV/PBO, neutralizes the final score against available style proxies, and stresses capacity and survivorship. The result is mixed: hold-out economics are better than momentum, and the same-target DSR is encouraging, but the full-grid DSR and PBO still force an exploratory interpretation (Table 4). The capacity, survivorship-haircut, and factor-neutral rows are sensitivity transforms of the same selected $N = 35$ overlay return stream; they bound fragility but do not provide independent confirmation.

Table 4: P0 robustness audit for the Route B 30-day residual momentum overlay. DSR deflates for the expected best searched result; PBO is CSCV probability of backtest overfitting. The full-grid DSR is a conservative heterogeneous-search approximation, not an exchangeability claim.

Check	Result	Interpretation
Test Sharpe before deflation	0.987	Better than 30d momentum's 0.740, before selection correction.
Paired overlay − momentum test	$t = 0.25$; corr 0.93	Return increment is not statistically distinguishable; improvement is risk dampening.
Bootstrap Sharpe 95% CI	[0.019, 1.796]	Positive but wide.
DSR, same-target trials	0.814	Best-case same-target framing is still below a 0.95 pass threshold.
DSR, full grid trials	0.003	Conservative bound; 0.987 Sharpe is below the 884-row null of 1.512.
CSCV PBO	50.0%	High but low-precision overfit risk across 28 strategies and 20 splits.
Fama–MacBeth signal t-stat	0.05	No convincing independent residual-alpha coefficient.
Factor-neutral Sharpe, return, FM t	1.006; 8.27%→5.17%; 0.05	Sharpe is mechanically similar because the residualized book remains momentum-like; FM t is the decisive orthogonality test.
Capacity Sharpe at \$100M / \$1B	0.948 / 0.864	\$100M is plausible; \$1B is extrapolated beyond the 1% ADV impact calibration.
PIT inclusion sensitivity	Mom. 5d 0.620→0.461, CI $[-0.49, 1.43]$; 20d 0.828→0.513; 30d 0.741→0.446	Partial inclusion-side check weakens the endorsed benchmark; overlay 30d is 0.987→0.725 and remains exploratory.
Survivorship stress-haircut Sharpe	0.837	Sensitivity does not destroy the result, but remains uncorrected.

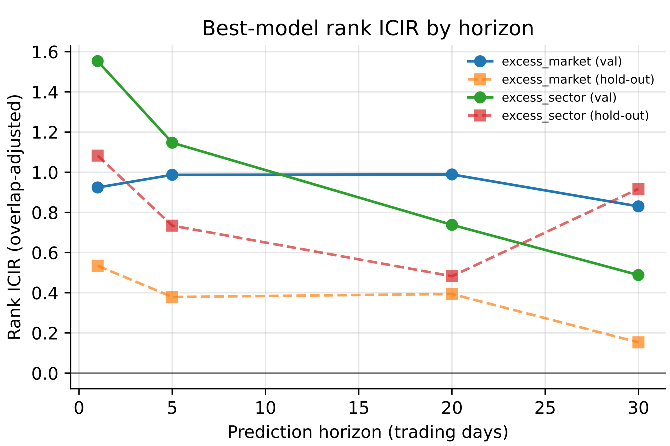

Figure 3: Best-model rank ICIR by horizon (validation vs hold-out), per target family.

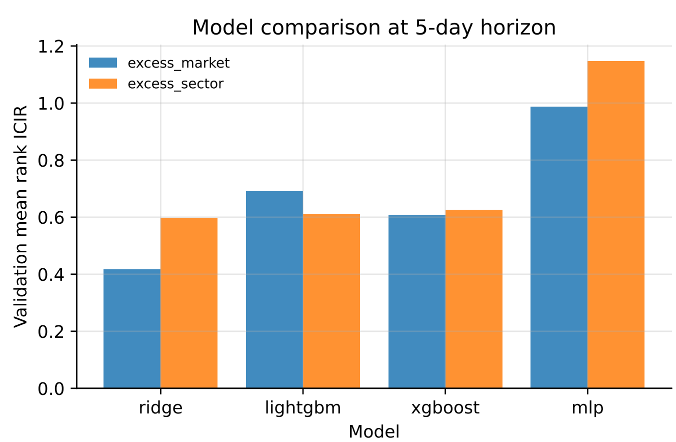

Figure 4: Validation mean ICIR by model at the 5-day horizon.

Overlap correction matters. Fig. 5 contrasts the overlap-adjusted hold-out ICIR with the naive overlapping value. The naive

Table 5: Single-evaluation hold-out (2022–2026), net of 5 bps/side cost and gross of stock-borrow. Model rows are validation selections; parenthetical ICIR is the naive overlapping value. Momentum Sharpe CIs are 5d $[-0.32, 1.63]$, 20d $[-0.12, 1.90]$, and 30d $[-0.22, 1.85]$; PIT-filtered Sharpes are 0.461/0.513/0.446 with CIs $[-0.49, 1.43]$, $[-0.45, 1.58]$, and $[-0.52, 1.50]$. Small PIT-vs-canonical Sharpe differences (0.620/0.828 vs 0.627/0.822) reflect date/universe filtering, not model selection. The PM decision row is 5d momentum; other horizons are sensitivity evidence, and the 30d block is too thin to rank statistically.

Horizon	Config	Rank IC	ICIR (raw)	Net Sharpe	Ann. Ret	Turnover
5d	Momentum baseline (sector)	**0.023**	**1.02** (2.05)	**0.63**	**5.9%**	**0.22**
5d	MLP *(val-selected, sector)*	0.013	0.73 (1.72)	0.17	1.2%	0.59
5d	Ridge (sector)	0.010	0.40 (0.99)	0.53	5.1%	0.65
5d	LightGBM (sector)	0.009	0.20 (1.17)	0.24	1.7%	0.53
20d	Momentum baseline (sector)	**0.031**	**0.58** (–)	**0.82**	7.8%	**0.36**
20d	LightGBM *(val-selected, sector)*	0.021	0.48 (2.42)	0.65	5.3%	0.66
20d	Ridge (sector)	0.017	0.34 (1.66)	0.83	9.0%	0.60
30d	Momentum baseline (sector)	0.028	0.70 (3.04)	0.74	7.8%	0.40
30d	MLP *(broad-grid val-selected, sector)*	0.027	0.92 (3.88)	0.69	6.5%	0.70
30d	LightGBM (not a result; val-rejected; below 1.512 null)	0.036	0.80 (4.34)	1.23	13.0%	0.63
30d	Route B overlay, $\lambda = 0.2$	0.028	0.76 (3.07)	0.99	8.3%	0.49

metric inflates sharply with horizon (to ≈3–10 at 30–90 days) while the adjusted metric stays modest (≤ 1.3); reporting the naive number would manufacture a long-horizon "signal" that is pure window overlap. The Sharpe>3/ICIR>5 sanity gate flags none of the adjusted runs.

Stability across regimes. Fig. 6 shows per-fold ICIR across the expanding walk-forward. The signal is positive but *variable* across folds, and the hold-out value (dashed) sits below the validation mean — the expected out-of-sample decay. The 90-day horizon is excluded from selection (too few non-overlapping validation periods for a stable ICIR), and the longer-horizon hold-out Sharpe values are estimated from few non-overlapping periods and should be read as noisy. The ensemble search also emits regime slices whenever a split overlaps the default stress windows (GFC 2008, COVID-2020, and inflation-2022); with the fixed ensemble split used here, the available ensemble stress rows are COVID-2020 in validation and inflation-2022 in test, while 2008 is covered only by the broader walk-forward stability analysis.

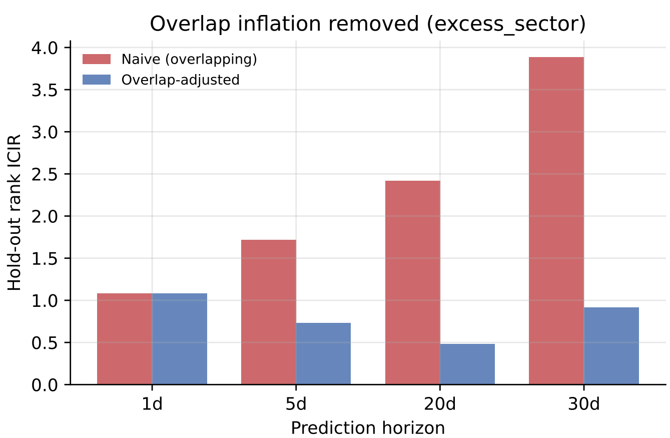

Figure 5: Overlap-adjusted vs naive (overlapping) hold-out ICIR by horizon.

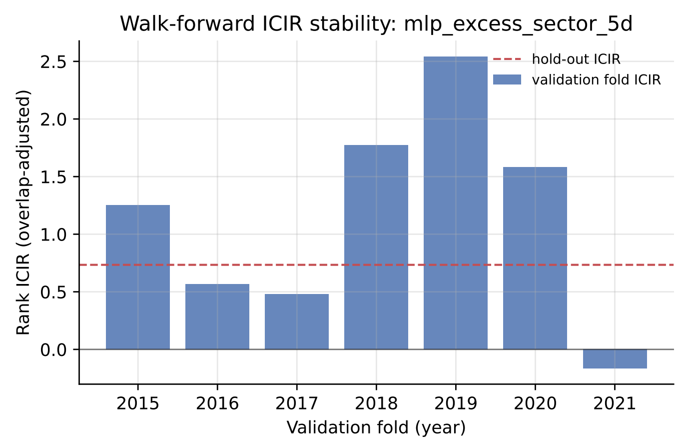

Figure 6: Walk-forward per-fold ICIR for the short-horizon model-comparison run.

Hold-out performance. Table 5 reports the single-evaluation post-2022 hold-out, net of 5 bps/side cost. The 5d and 20d blocks show the original conclusion: realistic cross-sectional equity "alpha" from price/volume features alone is small, and the trained models' apparent in-sample edge does not translate cleanly into out-of-sample value. The added 30d block is included to avoid horizon-selective framing: plain 30d models have attractive point estimates, but their longer-horizon sample is small and they have not passed the Route B deflation/attribution audit.

6 Discussion

Overfitting and the validation–hold-out gap. The clearest evidence of overfitting is in the results themselves: models selected on validation ICIR fail to beat a single momentum factor on the original 5-day rank and turnover benchmark, and the model with the best validation ICIR (the MLP) is not the best economic hold-out model. The final 30-day Route B rerun improves the momentum book only after the ML component is constrained to a residual sleeve, and even then the paired increment test shows no statistically distinguishable return addition. Standalone residual models still have weak economics and high turnover. We use heavy regularization (Ridge penalty, tree depth/leaf/subsample limits, MLP dropout + weight decay + early stopping), select hyperparameters on validation folds only, and report a single-evaluation hold-out precisely so this gap is *visible* rather than hidden,

Table 6: Evidence ladder for the endorsed 5-day momentum benchmark. The IC evidence survives better than the portfolio economics; the PM use is therefore benchmark/sleeve research, not standalone deployment.

Layer	Evidence	PM read-through
Signal existence	Non-traded 1d hold-out Rank IC CI $[0.0115, 0.0297]$; NW-21 $t = 4.39$. Traded 5d block-bootstrap IC $t = 2.41$, CI $[0.004, 0.042]$.	Best-supported evidence that the raw cross-sectional momentum signal exists; 5d is the traded benchmark.
Portfolio	Net Sharpe 0.62–0.63, mean-return $t = 1.31$, Sharpe CI $[-0.32, 1.63]$, turnover 0.22.	Corroborating economics, but not statistically tight standalone Sharpe.
PIT inclusion	Horizon-matched Sharpe $0.620 \rightarrow 0.461$; PIT-filtered CI $[-0.49, 1.43]$.	Inclusion-side bias is adverse here; deleted-name short-leg effects keep the net survivorship sign unresolved.
Borrow/capacity	25/50/100 bps borrow lowers Sharpe to 0.614/0.600/0.574; turnover remains lowest among tested books.	Still a useful low-turnover benchmark, gross-of-borrow point estimates are upper bounds.
Regimes	5d Sharpe by calendar slice: -0.95 (2008–11), 0.33 (2012–16), -0.14 (2017–21), 0.62 (2022–26).	Recent hold-out is momentum-friendly; no multi-regime funding proof.

with the Sharpe>3/ICIR>5 gate catching anything "too good".

Multiple testing. The original grid is large enough that the best Sharpe must be treated as a selected maximum, not as a single pre-specified experiment. For the Route B 30-day overlay, raw hold-out Sharpe is 0.987 and the bootstrap 95% Sharpe interval is $[0.019, 1.796]$. The same-target DSR is 0.814 within the 14-trial 30d target search. As a conservative search-process bound, full-grid DSR falls to 0.003 because the 884 archived searches imply an expected-max null Sharpe of 1.512, above both the overlay Sharpe 0.987 and the non-selected LightGBM 30-day Sharpe 1.226. That LightGBM row is not disproved, but it was not validation-selected and lacks a matching DSR/PBO audit, so it is not deflated alpha evidence. I report the 14-trial DSR as the cleaner like-for-like sleeve diagnostic and the 884-trial DSR as a conservative bound; the latter pools heterogeneous horizons and targets, inflating the penalty rather than proving a formal family-wise result. CSCV PBO is 50%, but only across 28 compact strategies and 20 splits, so it is an adverse instability flag rather than a precise probability. This is the most important statistical conclusion: the overlay is a materially better research lead than the old graph/MLP direction, but still not a statistically deflated production alpha; it fails a 0.95 DSR threshold under both the same-target and full-grid trial counts. The paired overlay-minus-momentum test reinforces that conclusion: the overlay is 93% correlated with momentum, reduces period volatility, and has a return-difference t-stat of only 0.25. Symmetric rigor also weakens any overconfident reading of the benchmark portfolio itself: the 5-day momentum book is retained as a discovery-to-hold-out survivor and PM benchmark, not as a post-hoc familywise significant portfolio Sharpe. Its native 1-day IC confirmation is strong even under a rough 12-signal by 6-horizon scan penalty, but the traded 5-day IC and especially portfolio Sharpe should be read as confirmatory and directional rather than a fully deflated search winner. The 5-day, 20-day, and 30-day momentum Sharpe point estimates are positive but imprecise, with mean-return t-stats of 1.31, 1.69, and 1.51 and bootstrap Sharpe CIs that all cross zero. The separate factor-sleeve book is even weaker: after charging the 12 pre-registered single-factor trials, the combined momentum/reversal book has DSR 0.055 and a bootstrap Sharpe interval that crosses zero. It is useful as a falsification exercise, not as a new product claim.

Route B reproducibility. The synchronized Route B artifacts close the largest traceability gap: the 30d overlay metrics can now be traced from local CSV/parquet files rather than only from a remote run note. A focused local rerun of the 30d momentum baseline now matches the Route B baseline row, so the absolute economic comparison in Table 3 is reproducible under one canonical evaluator. The frequency-scan and sleeve-book diagnostics also write lightweight CSV snapshots under `report/artifacts/`. The reason to keep Route B exploratory is not provenance; it is weak independent attribution, wide confidence intervals, and the full-grid DSR/PBO charge.

Factor attribution. The dataset lacks a full commercial risk model and fundamental value/quality factors, but we can still neutralize the final signal against available style proxies: liquidity/size (`log_dollar_volume`), volatility (`vol_20d`), beta (`beta_60d`), Amihud illiquidity, short-term reversal (`ret_5d`), and 12–1 momentum. The Route B overlay's score-neutralized Sharpe remains near 1.0, but that is not evidence of orthogonal alpha by itself: annual return falls from 8.27% to 5.17%, turnover rises from 0.49 to 0.70, and the non-overlapping Fama–MacBeth signal coefficient has only $t = 0.05$. The evidence therefore supports an economically interesting residual sleeve, not a clean claim of independent style-neutral alpha.

Look-ahead leakage. Addressed structurally: one-day execution lag, label purge on both sides, horizon embargo, and train-only preprocessing. We also verified that a no-embargo diagnostic inflates metrics, confirming the controls bind.

Survivorship bias. The dataset is current-membership-on-history, so the backtest is not point-in-time and cannot be treated as production evidence. The available `date_added` field supports an inclusion-side sensitivity: 72 current members were added on or after 2022-01-01, and after the baseline controls roughly 61–62 such symbols appear in hold-out rows that a PIT inclusion filter removes. Recomputing the sector-neutral momentum baseline with `sp500_member_asof==1` reduces Sharpe from 0.620 to 0.461 at 5d, from 0.828 to 0.513 at 20d, and from 0.741 to 0.446 at 30d. The 5d annual net return falls from 5.86% to 3.93%, max drawdown worsens from -14.6% to -17.3%, and the PIT-filtered 5d Sharpe bootstrap interval is $[-0.49, 1.43]$. That interval crosses zero even after the optimistic inclusion-side-only filter, so portfolio Sharpe is a directional result; the cleaner evidence for keeping the book is the 5d traded-horizon IC interval. The PIT-filtered 20d and 30d momentum Sharpe intervals are also wide at $[-0.45, 1.58]$ and $[-0.52, 1.50]$. Applying the same 30d filter post-training to the selected Route B prediction file and rebuilding the overlay score reduces Sharpe from 0.987 to 0.725, annual net return from 8.27% to 5.69%, and worsens max drawdown from -7.3% to -10.0% (`report/artifacts/pit_inclusion_sensitivity/`; `_5d/`; `_20d/`). This is an important hit to the point estimates, not a new validation pass. The audit also adds a delisting-side parametric sensitivity: applying annual delisting rate $\times$ average delisting loss to half the book leaves the Route B overlay Sharpe at 0.951 in a base case (2% annual delisting rate, -30% average delisting return) and 0.837 in a stress case (5%, -50%). Neither exercise is a true correction: the inclusion filter is directional rather than a bound, and deleted-name short-side exposure cannot be reconstructed from the current files. The likely direction is still adverse for a momentum book because missing bankrupt and delisted losers would tend to enter the prior-loser short leg. The right fix is point-in-time constituents and delisting returns, not a larger model.

Kronos and survivorship bias. Kronos is not reported as alpha evidence. Its only role here is a future route for embeddings on a point-in-time universe with delisted, acquired, and index-deleted names, which could reduce the current constituent-history survivorship problem.

Data quality. Prices are split-adjusted; dividend adjustment is uncertain and treated as a minor caveat. We winsorize features and filter the illiquid decile so bad ticks and microcaps do not dominate.

Turnover, capacity, and market impact. We charge 5 bps per side on realized turnover and rebalance only every h days. Turnover is itself decisive: the trained residual models churn ∼75–85% per rebalance, while the Route B overlay turns over 49.2% versus 40.1% for the 30-day momentum benchmark. A simple square-root impact stress test, calibrated to 10 bps at 1% ADV participation, gives overlay Sharpe 0.948 at \$100M AUM, 0.900 at \$500M, and 0.864 at \$1B, with 95th percentile participation rising to 25.3% at \$1B. The stress holds turnover fixed, so it is not a feasible execution policy once participation is high; linear scaling of p95 participation puts a 10%-ADV ceiling near \$395M. The \$500M/\$1B rows are extrapolated beyond the calibrated impact range, not capacity clearance; a real mandate needs participation caps and broker-calibrated execution. Stock borrow is not in the main evaluator, so the headline net figures should be read as gross-of-borrow upper bounds. A reproducible sensitivity (`report/artifacts/borrow_sensitivity/`) deducts 25/50/100 bps annual borrow on 50% short notional: 5d momentum Sharpe falls from 0.627 to 0.614/0.600/0.574, 20d momentum from 0.822 to 0.809/0.796/0.769, 30d momentum from 0.740 to 0.728/0.716/0.693, and the 30d Route B overlay from 0.987 to 0.972/0.957/0.927. These haircuts do not change the ranking, but they reinforce that the paper is a research benchmark rather than a trade-ready long–short implementation.

Regime dependence. Walk-forward folds span 2008, 2020, and 2022 stress periods; Fig. 6 shows the signal is not uniform across regimes, motivating the market/breadth/dispersion regime features rather than a single static model. A focused calendar-slice diagnostic (`report/artifacts/momentum_regime_diagnostics/`) makes the economic fragility explicit for the no-fit momentum book: 5d Sharpe is -0.95 in 2008–2011, 0.33 in 2012–2016, -0.14 in 2017–2021, and 0.62 in 2022–2026; the analogous 30d Sharpes are -0.91, 0.20, -0.12, and 0.67. These are calendar-slice stresses, not independent retraining or CPCV proof, and saved Route B predictions do not support a pre-2022 overlay Sharpe distribution. The frequency-band diagnostic makes the same point at the factor level: Amihud illiquidity is a strong discovery-period signal but dies in the 2022+ hold-out, while momentum is weak in parts of discovery and strong in the recent hold-out. This is why single-period IC surfaces are treated as hypothesis generators, not proof.

Economic intuition. The best-supported raw signal out of sample is cross-sectional momentum — a well-documented behavioural underreaction effect — which is why the single-factor baseline is hard to beat. The support is strongest as a signal-existence result; traded-horizon IC is positive at 5–10 days and marginal by 20–30 days, so sizing should not lean on recent Sharpe alone. The sleeve-book diagnostic confirms that a theoretically attractive momentum-plus-reversal product is not enough by itself: reversal is mildly orthogonal, but too weak and too high-turnover to improve hold-out Sharpe. The Route B result is useful because it gives the model a narrower economic job: find residual structure that can be added to, not substituted for, momentum, and then survive DSR rather than merely improve a point estimate. The current Route B improvement is therefore best read as a candidate risk-timing effect, not as established residual alpha.

7 Limitations and Conclusion

This study delivers an honest, reproducible S&P 500 cross-sectional relative-return pipeline using only price/volume and light metadata. The key discipline is methodological: non-overlapping evaluation, walk-forward selection, explicit leakage controls, and a single-factor baseline that every model must clear. The realistic conclusion is sobering: broad multivariate, graph, MLP, and residual-overlay searches have not demonstrated production-grade alpha. Momentum is the only raw factor with traded-horizon IC support and corroborating 1-day IC confirmation, and the 5-day sector-neutral implementation is the benchmark I would keep because its traded-horizon IC interval is positive and turnover is lowest (Sharpe 0.62–0.63, turnover 0.22, block-bootstrap IC $t = 2.41$). This is not a production-ready endorsement: mean return $t = 1.31$, bootstrap Sharpe CI $[-0.32, 1.63]$, PIT-filtered Sharpe 0.461 with CI $[-0.49, 1.43]$, negative or weak pre-2022 slices, gross-of-borrow accounting, and high-AUM capacity extrapolation all keep the recent Sharpe from being capital-allocation evidence.

My PM decision would be to keep the 5-day momentum book as the live research benchmark and small-capacity paper sleeve, size any real exposure off IC stability and turnover rather than headline Sharpe, and require a true point-in-time universe with delisting returns before funding it. Route B is a 30-day risk sleeve, not an improvement claim on that 5-day benchmark; it earns a next-round budget only if it passes CPCV-style stability, a 5-day horizon-matched overlay check, and a stronger style-neutral return test. Its 2022–2026 point estimate improves the 30-day momentum book's Sharpe from 0.740 to 0.987 and drawdown from -13.8% to -7.3%, but the paired increment is not statistically distinguishable ($t = 0.25$), correlation with momentum is 93%, canonical Rank IC is 0.0279 versus momentum's 0.0282, full-grid DSR is 0.003, PBO is 50%, and Fama–MacBeth $t = 0.05$. A naive vol-timing control also fails to match the drawdown improvement, so Route B is not dismissed as mere vol-targeting, but it remains exploratory. The useful output is the triage: keep 12–1 momentum as the core benchmark, treat residual ML as a small risk/drawdown research sleeve, and make DSR/PBO, PIT survivorship, capacity, and turnover part of model selection rather than post-hoc decoration.